

ITDxDDA Diploma in Immersive Media Year 2 (2024/25) Semester 2.2	Version 1
	Week 3 - 8
Assignment Project 1 (35%) Design & Development of an Augmented Reality experience	

OVERVIEW

Goal

To create an Augmented Reality Mobile Application to provide interaction points for:

- Engaging consumers with a [Singapore Heritage Brand](#).
 - OR
- Location based AR Notes
 - OR
- Nurturing a Virtual Avatar

This is a pair-work project.

Register your pairings here: [DDAxITD_Asg1 Groupings.xlsx](#)

Objectives

The Augmented Reality Application must be a full experience **(must have a clear start and clear end to it)** and be able to fulfill **at least** one of these conditions and the DDA portion :

- To provide an AR enhancement to informative content provided by existing physical content (ie. Printed brochures / posters / signage / textbook)
- To introduce gamification features / create a simple game by tapping on AR technology, related to the brand, product, or person.
- (DDA) To integrate Firebase services (database and authentication)

Example Scenarios

Here are some scenarios to help you get started. How would you plan or re-design these situations by tapping on AR:

1. A heritage brand has invested heavily in print commercials (posters, fliers, billboards, etc.) They do not want to reprint them and hope to use AR to introduce new life to these existing commercials.
2. Your school would like to promote some underutilized areas on campus by creating an AR campaign. The campaign will allow users to visit these areas and leave virtual notes or emoji reactions for others to view.
3. You are an avid collector of succulents and would like to spread the joy of growing and caring for succulents. You decide to create an AR experience where users can care for virtual succulents while learning more about them.

NOTE: These are just **EXAMPLES**. Feel free to pitch your own scenarios, but they should relate back to the broad theme of Personal/Commercial Branding or Education/Training.

GUIDELINES

(ITD) AR App Prototype

Your end product will be assessed based on the following criteria:

1. Technical production, smoothness of flow, thorough testing to ensure no bugs
2. Design and Usage of physical markers, choice of AR sensing
3. Visual, textual and audio cues used. Feedback mechanism.
4. Engagement value of prototype

You may want to ask yourself the following questions as you go along:

- Does the content being presented need an AR application?
 - If not, how can I supplement it with more information or make it easier to digest with AR?
- Consideration for physical space, physical lighting for scanning AR markers?
How can I make the experience seamless for my audience and the various devices
- Practical design considerations – whether prototype suits environment, physical space, group size, amount of time to learn and use the application
- Interactivity of application, am I using suitable typography, menu design, visual hints on using the AR app
- Balance of 2D and 3D elements
- For games, think about fulfilment of task. Visual and audio cues/indicators placing and timing
- Social sharing options or whether AR activities can involve more than 1 person (does not have to be in a multiplayer context)
- Rewards consideration for games. What kind of achievement rewards
- Load & waiting time. Visual and textual hints when there is no scanning or waiting for something to happen.
- Have I tested enough? User physical height? Is it at only certain angles my prototype works?
- How AR is used to enhance the experience? For 1 person? For a group setting?
- If there's audio, should you cue users to plug in headphones? Don't forget it's an open setting environment.
- Screen size and device limitation.
- Performance of application. Make sure devices do not overheat running your app.
- You may consider using APIs to enhance the experience.
- You may consider using databases or file system to allow storage of information and easy updating of new content materials

(DDA) Full Stack Dev

Your end product will be assessed based on the following criteria:

1. Technical production, smoothness of flow, thorough testing to ensure no bugs
2. Design and Usage of database, usefulness of web dashboard and front-end
3. Engagement value and integration of Firebase and Unity project

You may want to ask yourself the following questions as you go along:

1. What data structure should you adopt
2. What kind of content to display and interact on the dashboard
3. Useful administrative features for ease of real-time updates?
4. What kind of data to be stored in game, when is the data being stored?
5. Have I tested enough? User input testing? Back end data checks?
6. Completeness of project
7. User experience and flow of application with Firebase services

ASSIGNMENT WEIGHTAGE

ITEMS	WEIGHTAGE
Basic AR Application (MUST HAVES) ○ Application must work in Unity Editor. ○ Application must have a clear start (e.g. a start menu for users to start the experience) ○ Application must have a clear end (e.g. player is informed they have completed the experience and prompted to return to start screen) ○ Application must use Image Tracking. ○ Spawned objects from Image Tracking must be interactable through UI. ○ There must be an overarching objective for users (e.g. scan all image targets, collect all objects, etc.) ○ There must be some indication on player progress (e.g. UI on screen) ○ There must be some feedback when player has completed a task (e.g. collected 1/5 objects, scanned 1/3 targets, etc.) ○ Models used must not just be primitive shapes in Unity (e.g. Cube, Capsule, Cylinder, etc.) ○ 3D Models must be textured. Basic DDA portion (MUST HAVES) ● Application must be linked to Firebase Realtime database ● Application must have a sensible data structure design ● Appropriately linked up authentication services ● There must be some interaction with the Firebase Realtime database ● There must be some authentication interaction ● Database JSON file must be filled with sample data ● Interaction of in game with the database must be shown	40% <i>(Marks will only be awarded if ALL features are shown in demo video)</i>
Additional Features <u>Refer to Additional Features Checklist below.</u>	Max of 25% <i>(Marks will only be awarded for features shown in demo video)</i>
Overall Quality	15%
Peer Evaluation	20%
GRAND TOTAL	100%

ADDITIONAL FEATURE CHECKLIST (ITD)

AR Features

Feature Description	Marks	Done?
The application can be deployed to a mobile device.	2	
The application incorporates additional AR features, such as plane detection or face tracking.	3	
AR objects within the application can interact with one another.	5	
AR objects are aware when other AR objects appear.	5	
The application has various forms of interaction between the player and AR elements.	3	
The application can track multiple copies of the same image, with one object per image.	7	

UIUX

Feature Description	Marks	Done?
The design of menus is cohesive and consistent with the rest of the application.	3	
The application includes custom assets.	2	
UI elements are purposefully used and placed.	2	
The application considers mobile screen requirements.	3	

Game Juice

Feature Description	Marks	Done?
Audio effects are effectively implemented within the game.	3	
The game includes feedback mechanisms.	3	
The game responds appropriately to player actions.	3	

Promotional Material

Feature Description	Marks	Done?
Promotional Trailer for the project	3	
Promotional social media posts for the project	2	

ADDITIONAL FEATURE CHECKLIST (DDA)

DDA Features

Feature Description	Marks	Done?
The application can be deployed to a mobile device.	2	
The application incorporates Real-time database rules and indexes	3	
Handling of User Authentication issues	5	
A sensible related web related frontend that is deployed	3	
A web portion is made that can be of an extension to the application or an administration frontend that interacts with Firebase services in real-time	10	
Full-fledged end-to-end integration with Web + Unity + Firebase	5	

UIUX

Feature Description	Marks	Done?
The application shows user-friendly error messages	3	
The application has a seamless fallback mechanism for Database/Auth handling	3	
UI elements are purposefully used and placed.	4	
Responsive web design for web frontend	2	

Promotional Material

Feature Description	Marks	Done?
Promotional Trailer for the project	3	
Promotional social media posts for the project	2	

DELIVERABLES (ITD)

1. Github link to Unity Project
 - **Unity Project** must include all asset files (e.g scenes, scripts, materials, textures, packages, models, videos, etc)
 - Ensure all files are organised and have a proper file name
2. 1 compiled **Unity Project** (.apk/xcodeproj).
3. 1 **video demo** compilation of the whole AR experience.
 - Must showcase the start to end of the AR experience.
 - Can be reasonably sped up if needed (~1.2x)
 - Must showcase all features you have marked as completed
 - Must showcase how features have been developed and implemented
4. 1 set of **Read Me** & walkthrough
 - Readme should include detailed instructions (key controls, “game cheats/hacks”, answer key, etc) on how to use and run your application.
 - Indicate the platforms/hardware required to run your application
 - Indicate the limitations or bugs in the application
 - Indicate **references and credits to the models/textures/materials** used
 - If there are any games/application that requires solving, please write out your solutions.
5. 1 **Feature Checklist**
 - Must indicate which features in the checklist are completed.
 - A template will be provided on BrightSpace.
6. **Promotional Materials** (Optional) .zip file

FINAL SUBMISSION		
Item(s)	Naming Convention	Submission Channel
1	ITD_Asg1_GithubLink.txt	HTML link to github repository.
2	ITD_Asg1_ProjectName.apk/xcodeproj	The compiled build from Unity.
3	ITD_Asg1_ProjectName_Video.mp4	The video demo of your application.
4	ITD_Asg1_ProjectName_Walkthrough.pdf	The walkthrough of the application.
5	ITD_Asg1_FeatureChecklist.pdf	The feature checklist detailing completed features of the project.
6	ITD_Asg1_PromotionalMaterials.zip	A zip file containing the promotional materials for your project.

DELIVERABLES (DDA)

1, **Unity Project** must include all asset files (e.g scenes, scripts, materials, textures, packages, models, videos, etc)

- Ensure all files are organised and have a proper file name
- Ensure that appropriate .gitignore is done

2. Web Project files

- Proper file management and appropriately named files
- Firebase config is properly setup
- Additional bonus for website to be hosted externally

3. Video

- To record interaction of game and database in real-time
- Show case all needed key interactions with Firebase

4. Database Export

- Include all need sample data information

5. Readme

- The readme is to contain all documentation and research done.
- What content you are displaying
- What is the application catering for
- Documented Wireframes/game flow
- State & Attribute all external assets/libraries used
- State original artwork/assets done

7. 1 **Feature Checklist**

- Must indicate which features in the checklist are completed.
- A template will be provided on BrightSpace.

DELIVERABLES (DDA)

FINAL SUBMISSION		
Item(s)	Naming Convention	Submission Channel
1	DDA_Asg1_GithubLink.txt	HTML link to github repository. (if you have the web front, do include it too)
2	DDA_Asg1_ProjectName_Web.zip	Zip of web application (if done)
3	DDA_Asg1_ProjectName_Video.mp4	Video demo of the real-time database interaction with the application
4	DDA_Asg1_ProjectName_Database.json	JSON Export of Database with sample data
5	DDA_Asg1_Readme.pdf	Detailed readme of DDA features and content
6	DDA_Asg1_FeatureChecklist.pdf	The feature checklist detailing completed features of the project.

DUE DATES

Ideation	Week 4 Message your tutor on Teams individually showcase the POC of your idea for the assignment.
Final Submission	Week 8, Sunday (14/12/25), 2359hrs

LATE SUBMISSION

Late submission will be **penalised** (10% of the marks for each day late after 12 noon). Submission will not be accepted after 5 days (including weekends and public holidays) from the date of submission.

PLAGIARISM AND COPYRIGHT ISSUES

Plagiarism means, “*copying any part of a source, and then submitting it, claiming that it is your own work.*”

Please ensure that all the works submitted by you are not copied from other sources. Any attempt to plagiarize will be dealt with severely, and it may result in your failing the module.

If you have made any references to certain materials, make sure you cite the sources by acknowledging and providing the information necessary to find the source (e.g.

Title and author of book, Internet links, etc). This includes any code obtained from **Gen-AI sources and online tutorials**.